

Creating an Entire Doubly Linked List

```
#include <stdio.h>
#include <stdlib.h>

struct node {
    struct node* prev;
    int data;
    struct node* next;
};

int main()
{
    struct node* head = NULL;
    struct node* ptr;
    head = createList(head);

    ptr = head;
    while(ptr != NULL)
    {
        printf("%d ", ptr->data);
        ptr = ptr->next;
    }
    return 0;
}
```

```
struct node* createList(struct node* head)
{
    int n, data, i;
    printf("Enter the number of nodes: ");
    scanf("%d", &n);

    if(n == 0)
        return head;

    printf("Enter the element for the node 1: ");
    scanf("%d", &data);
    head = addToEmpty(head, data);

    for(i=1; i<n; i++)
    {
        printf("Enter the element for the node %d: ", i+1);
        scanf("%d", &data);
        head = addAtEnd(head, data);
    }
    return head;
}
```

NULL
head

n data i

```
printf("Enter the element for the node 1: ");
scanf("%d", &data);
head = addToEmpty(head, data);
```

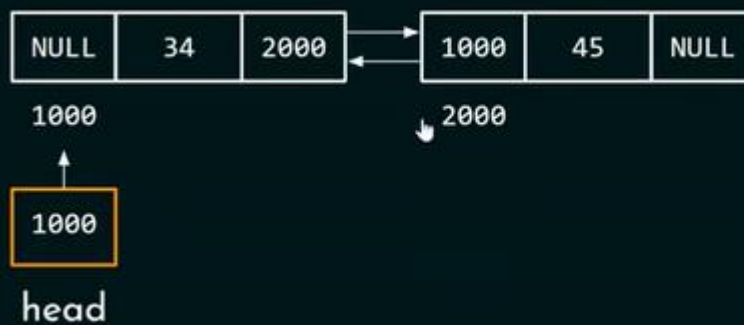


```
printf("Enter the element for the node %d: ", i+1);
scanf("%d", &data);
head = addAtEnd(head, data);
```

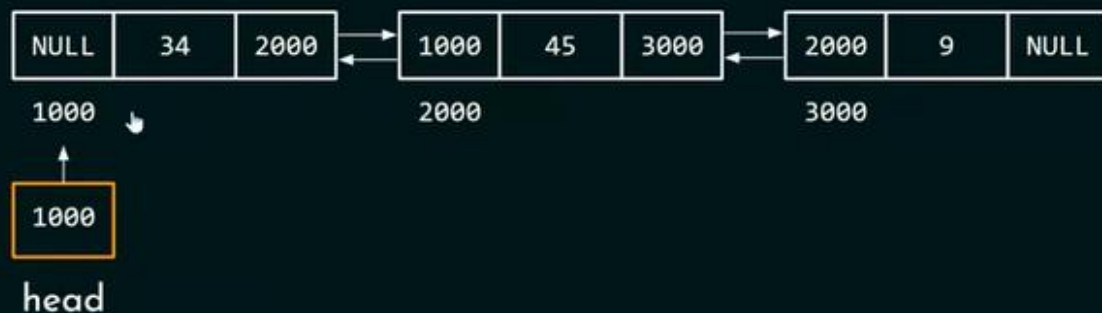
Console Window

```
Enter the number of nodes: 2
Enter the element for the node 1: 34
Enter the element for the node 2: 45
```

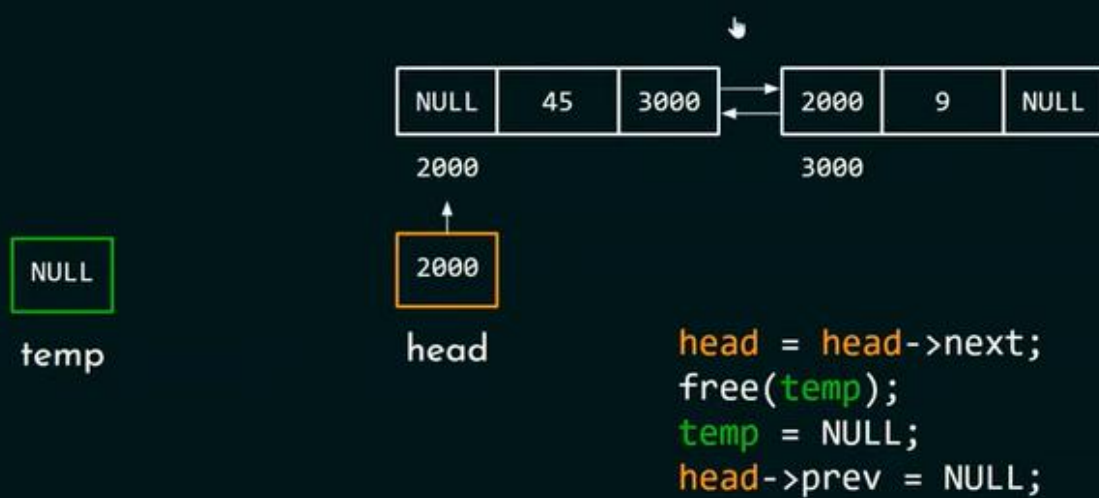
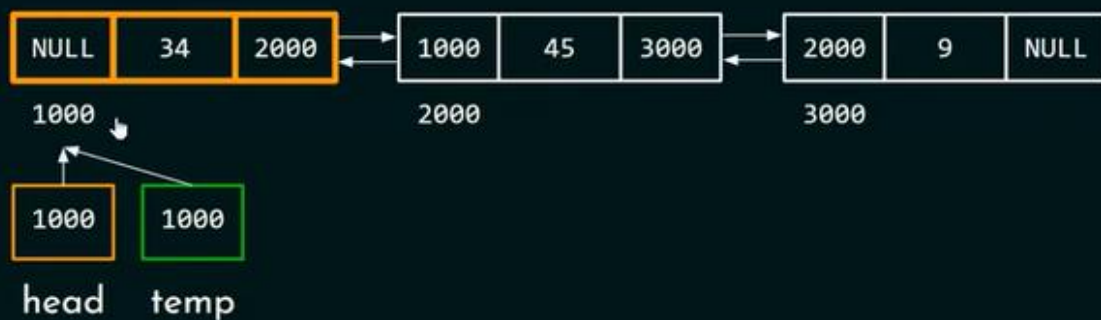
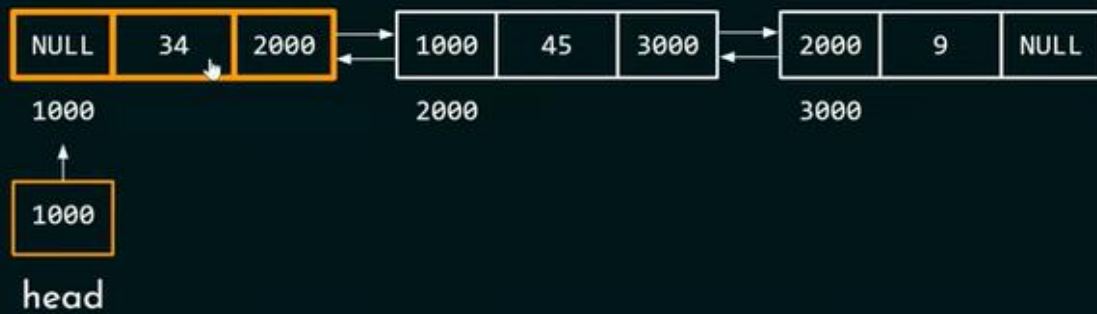
```
printf("Enter the element for the node %d: ", i+1);
scanf("%d", &data);
head = addAtEnd(head, data);
```



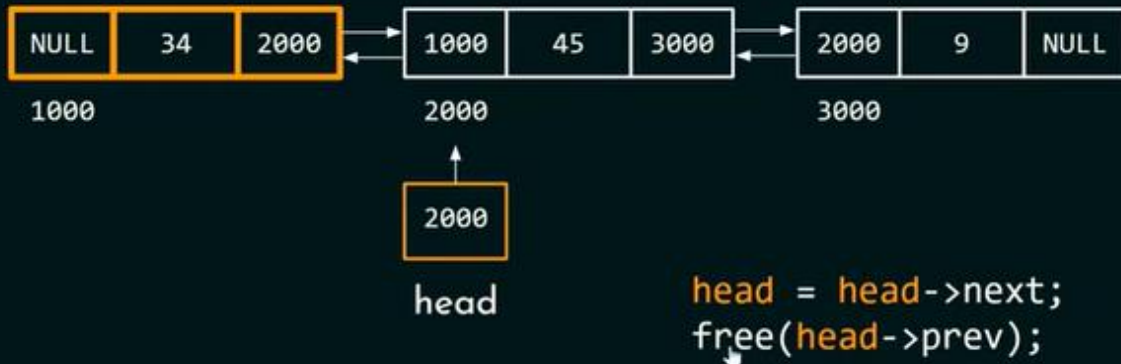
Doubly Linked List (Deleting the First Node)



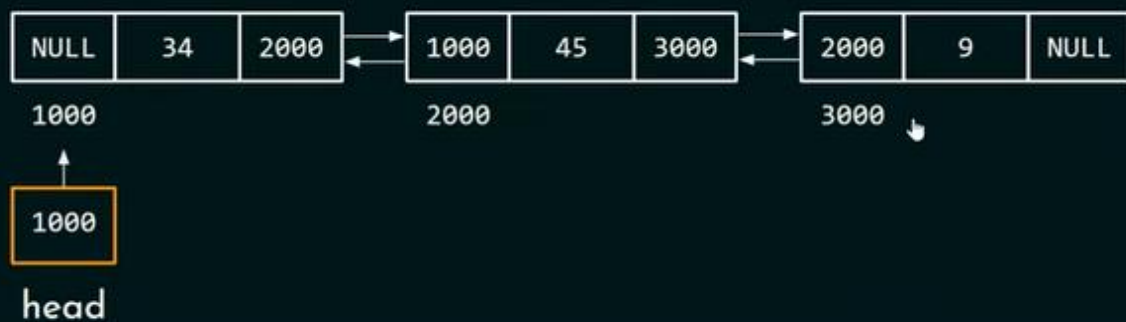
Our target is to delete this node

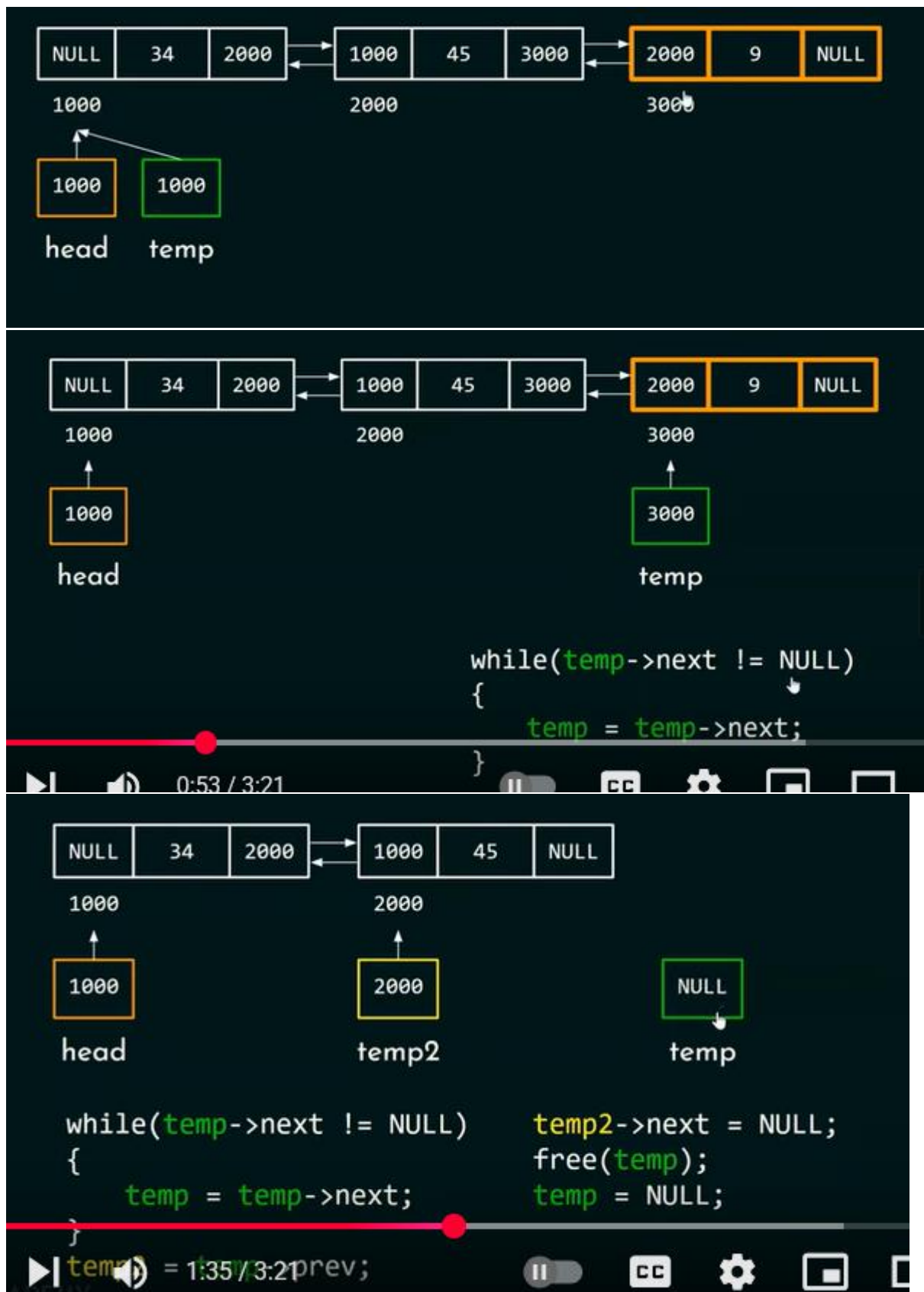


METHOD 2



Doubly Linked List (Deleting the Last Node)

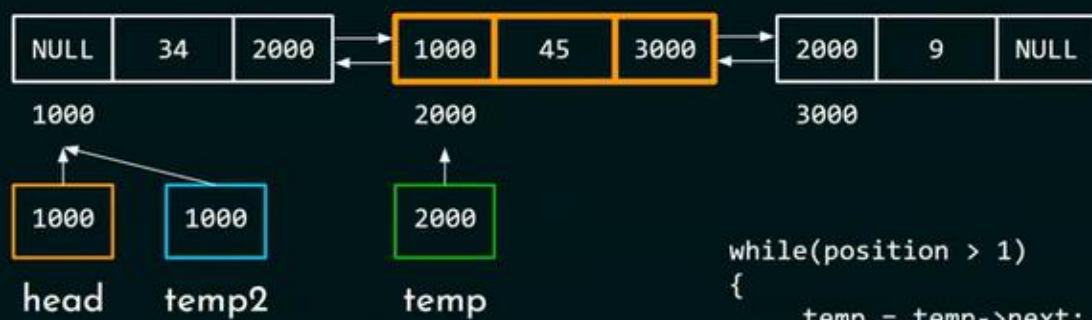




Doubly Linked List (Deleting the Intermediate Node)



position = 2



```
while(position > 1)
{
    temp = temp->next;
    position--;
}
temp2 = temp->prev;
```